

Lesson 9 Assignment Due Monday Oct 12, 2026

Instructions

Copy and paste SAS code into Canvas to submit your answers.

Problem 1

We have collected the following data on five subjects:

ID	AGE	GENDER	Grade Point Average (GPA)	College Entrance Exam Score (CSCORE)
1	18	M	3.7	650
2	18	F	2.0	490
3	19	F	3.3	580
4	23	M	2.8	530
5	21	M	3.5	640

- (a) Write the SAS statements necessary to create a SAS dataset.
- (b) Add the statement(s) necessary to compute the mean grade point average and mean college entrance exam score.
- (c) We want to compute an index for each subject, as follows:

$$\text{INDEX} = \text{GPA} + 3 \times \text{CSCORE}/500$$

Modify your program to compute this INDEX for each student and to print a list of students in order of increasing INDEX. Include in your listing the student ID, GPA, CSCORE, and INDEX.

Problem 2

Given the following set of data:

Social Security Number	Annual Salary	Age	Race	Hire Date
123874414	28,000	35	W	4/12/1971
646239182	29,500	37	B	3/15/1992
012437652	35,100	40	W	4/15/1983
018451357	26,500	31	W	12/10/2002

- (a) Create a SAS data set using the data above. Compute the average annual salary and average age. **NOTE:** Enter the annual salary values into DATALINES with commas.
- (b) If all subjects were in a 30% tax bracket, compute their taxes (based on annual salary) and print out a list, in Social Security number order, showing the annual salary, the tax, and the hire date.

Problem 3

- (a) Import the dataset https://manctsui.github.io/datasets/billboard_2021_01.csv and sort data by weeks on board.
- ~~(b) To this dataset, write in a new column called peak_position which is a 1 if the rank equals peak rank and a 0 otherwise.~~